



CSDE 502

Ethics & Norms

April 4, 2025



Course Structure & Expectations

- This is a largely discussion-based course, most of the assignments involve coming to class prepared to discuss readings by
 - doing them! and
 - coming prepared with questions or topics you want to discuss



Course Assignments

- Assignments and course are credit/no credit
- Assignments include:
 - Discussion participation
 - CITI Training
 - Final Exam (Take home, written)



Thinking Ethically

The Utilitarian Approach

- greatest balance of good over evil
- What are the possible actions?
- Who will benefit and who will be harmed?
- Which actions allow greatest benefits and least harm?

The Rights Approach

- individual's right to choose for themselves
- right to truth
- right to privacy
- right not to be injured
- right to what is agreed

The Fairness or Justice Approach

- How fair is an action?
- Does it treat everyone in the same way?
- Does it show favoritism and discrimination?

The Common-Good Approach

- Community members bound by pursuit of common values and goals
- the social policies, social systems, institutions, and environments on which we depend are beneficial to all

The Virtue Approach

- assumes there are certain ideals toward which we should strive
- What will promote the development of character within myself and my community?



Fostering Integrity, Ch. 2: Core Values

- Objectivity
- Honesty
- Openness
- Accountability
- Fairness
- Stewardship



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Objectivity

Karl Popper (1999)

- 1) pose refutable hypotheses,
- 2) test hypotheses with relevant evidence,
- 3) state the results clearly & unambiguously to any interested person

Goal is reproducibility

Best intentions not always sufficient!



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Honesty

- Starting assumption for institutions and stakeholders
- Forms of dishonesty
- plagiarism & data fabrication
- p-hacking, cherry-picking, misrepresentation of results (bad figures, bad interpretations, bad headlines)
- Non-reporting



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Openness

- Transparency & presenting ALL the information relevant to a decision
- Sometimes there is a pull between adherence to the value of openness and other goals e.g. commercial gain, subject privacy



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Accountability

- Individual accountability
 - obligation to others in “web of science” and society
- Mutual accountability
 - Peer to peer
 - Mentor to mentee
 - Institutions to individuals



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Fairness

- Appropriate and ANNOUNCED criteria
- Authorship, citation, acknowledgments
- Human and living subjects
- Society



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Stewardship

- Attending to professional relationships & curating working environments
- Service to societies and institutions
- Mentorship and education
- Society



Mertonian Norms & Counter-Norms

A normative system is:

- a systemic or societal attribute
- the set of all norms associated with a particular social system
- the members' collective subscription to the norms
- the members' collective weighting of the norms' importance and applicability

A normative orientation is:

- an individual attribute
- describes a unique pattern of subscription and resistance to a *normative system*



Mertonian Norms & Counter-Norms

- Robert Merton came up with four **research norms** in 1942 and acknowledged there were direct **counter-norms** to each
- A **norm** may be a behavior that is typical within the social group, OR a behavior that is deemed desirable or ideal for the social group
- Ian Mitroff (1974) outlined four **direct counter-norms** to Merton and Michael Mulkay (1976, 1980) argued neither one describe a **normative system** actually adhered to by the scientific community
- Anderson et al. (2010) propose 4 more pairs of **norms and counter-norms** and study their adherence in modern science



Mertonian Norms & Counter-Norms

Norms

1. communalism
2. universalism
3. disinterestedness
4. organized skepticism
5. governance
6. quality
7. calling
8. breadth

Counter-norms

1. individualism
2. particularism
3. self-interestedness
4. organized dogmatism
5. administration
6. quantity
7. employment
8. narrowness



Communalism vs Individualism

- **Communalism:** Scientists openly share new findings with colleagues.
- **Secrecy/Individualism:** Scientists protect their newest findings to ensure priority in publishing, patenting, or applications.



Universalism vs Particularism

- **Universalism:** Scientists evaluate research only on its merit, i.e., according to accepted standards of the field.
- **Particularism:** Scientists assess new knowledge and its applications based on the reputation and past productivity of the individual or research group



Disinterestedness vs Self-interestedness

- **Disinterestedness:** Scientists are motivated by the desire for knowledge and discovery, and not by the possibility of personal gain.
- **Secrecy/Individualism:** Scientists compete with others in the same field for funding and recognition of their achievements.



Organized skepticism vs Organized dogmatism

- **Organized skepticism:** Scientists consider all new evidence, hypotheses, theories, and innovations, even those that challenge or contradict their own work.
- **Organized dogmatism:** Scientists invest their careers in promoting their own most important findings, theories, or innovations.



Governance vs Administration

- **Governance:** Scientists are responsible for the direction and control of science through governance, self-regulation and peer review.
- **Administration:** Scientists rely on administrators to direct the scientific enterprise through management decisions.



Quantity vs. Quality

- **Quality:** Scientists judge each others' contributions to science primarily on the basis of quality.
- **Quantity:** Scientists assess each others work primarily on the basis of numbers of publications and grants.



Calling vs. Employment

- **Calling:** Scientists view science as serving a purpose worth of personal sacrifice.
- **Secrecy/Individualism:** Scientists work in accordance with the terms of their employment, such as pay, benefits, working hours, and vacation time.



Breadth vs Narrowness

- **Breadth:** Scientists fulfill a broad range of responsibilities in the areas of teaching, research and service.
- **Narrowness:** Scientists put more of their time and effort into their research than into any other aspect of their work.

